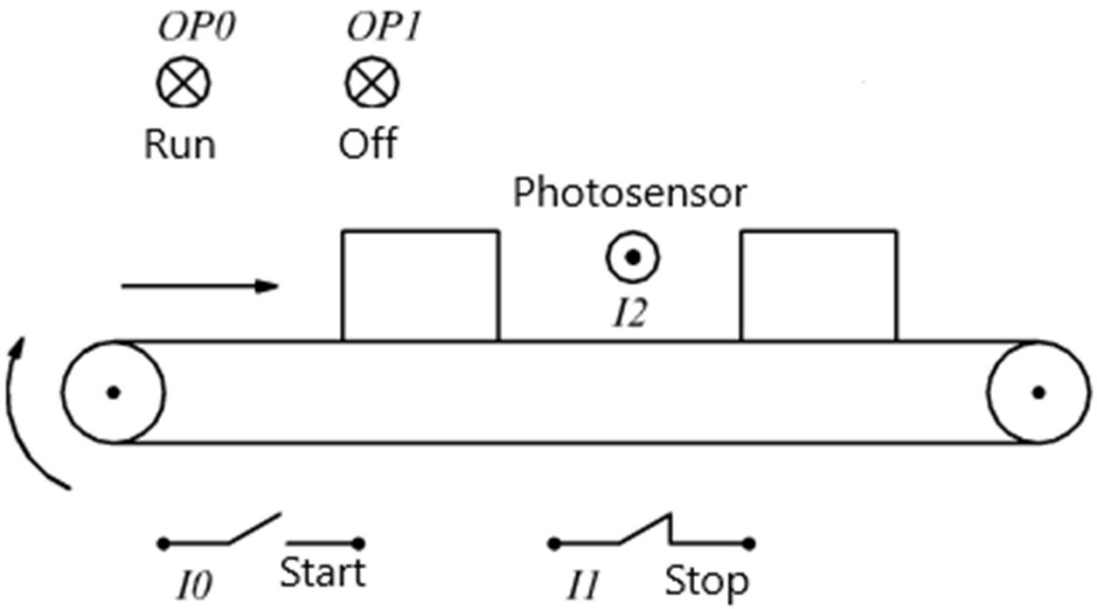
System Overview & Requirements



Create a piece-counting system with a conveyor belt. The system consists of a conveyor belt driven by a pneumatic actuator through two start and stop buttons, the start and stop status must be detected by two distinct lamps.

The pieces to be counted are placed on the belt and each time one of them passes in front of the connected photocell it must be counted.
The belt can be stopped at any time by pressing the stop button: it can be restarted by pressing the start button again, but piece counting must resume from where it was interrupted.

Assignment

Develop the automatic control to implement the task using Automation Studio programming, using two separates coding:

- Ladder Code with Step-Transition method and PLC cards;
- Grafcet code.

Each of the two code must be well-organized, with the use of subroutines (FC, Partial GRAFCETs,…). All movements of the plant shall be actuated using pneumatic cylinders controlled by proper electro valves.

Moreover, develop two clear and understandable HMI interfaces in Automation Studio with: Start, Stop, Emergency, alarms, counting, actuators and all system animation… (movements and visibility) and all sensors, counters and pilot lights needed to simulate the real plant.